

Perform Web Server Reconnaissance Using Various Tools

Execution mode	Authorized, isolated training lab; no public systems were targeted.
Output status	Representative/sanitized lab output recorded for submission. Replace with instructor-required screenshots if a live lab run is performed.

Aim

Identify the web server type, version, exposed ports, technology stack, and common misconfigurations in the authorized lab server.

Theory

Web-server reconnaissance collects service banners and response metadata. The report uses a synthetic Apache server at 192.168.56.30. Findings are for training and are not evidence about any Internet host.

Tools and Lab Scope

- Nmap
- Nikto
- WhatWeb
- curl
- Metasploitable/DVWA-style lab server

Procedure and Recorded Execution

Step 1. Run service detection - Scan only the approved host and limited web ports.

```
nmap -sV -p 80,443,8080 192.168.56.30
80/tcp open  http Apache httpd 2.4.57
443/tcp closed
8080/tcp closed
```

Step 2. Run HTTP NSE scripts - Collect headers and page title from the same lab host.

```
nmap -p80 --script=http-headers,http-title,http-server-header 192.168.56.30
Title: DVWA Training
Server: Apache/2.4.57
```

Step 3. Inspect headers - Compare curl output with Nmap.

```
curl -I http://192.168.56.30
HTTP/1.1 200 OK
Server: Apache/2.4.57
Content-Type: text/html
```

Step 4. Fingerprint and scan - Run WhatWeb and Nikto against the lab server.

```
whatweb http://192.168.56.30
Apache[2.4.57], PHP[8.2], HTML5
nikto: missing Content-Security-Policy; server header disclosed
```

Web-server fingerprint output

The server was identified as Apache 2.4.57 with PHP 8.2 serving the DVWA training page. Nikto reported informational disclosure and missing hardening headers, not a confirmed exploit.

Item	Result
Open service	80/tcp HTTP
Web server	Apache httpd 2.4.57
Application clues	PHP 8.2; DVWA Training title
Nikto findings	Server banner disclosure; missing CSP; no dangerous default file in fixture
Confidence	High for service identity; version may be normalized by a proxy

Result

Reconnaissance completed. The fingerprint provides enough information to choose appropriate defensive configuration checks and version-management actions.

Safety and Submission Notes

- Rate-limit scans and restrict them to the approved IP.
- Do not treat a banner as a vulnerability by itself.
- Validate version findings from more than one source.

Viva Questions and Answers

Purpose of -sV?

It performs service/version detection by probing open services.

Why is Server useful to attackers?

It can narrow the set of version-specific weaknesses to investigate.

What does Nikto look for?

Known dangerous files, weak headers, outdated software, and common misconfigurations.

WhatWeb vs Nikto?

WhatWeb fingerprints technologies; Nikto checks web-server weaknesses and known files.

Why identify versions?

Accurate versions help defenders prioritize patches and reduce false assumptions.